

Sports Car Price Analysis and Market Trends

1. Data Sourcing

Dataset 1: sports_car_prices

The primary dataset contains detailed specifications and pricing for a wide range of sports cars from various manufacturers. It covers key performance variables including engine size, horsepower, torque, and 0–60 MPH acceleration time alongside market price, making it well-suited for regression and correlation analysis.

Dataset Details: 1008 Rows & 8 Columns

Size: 50 KB

Access: Downloaded directly from Kaggle as a CSV file.

Source: Kaggle – <https://www.kaggle.com/datasets/rkiattisak/sports-car-prices-dataset>

Dataset 2: car_price_analysis

A supplementary dataset containing general car pricing and specification data used to validate pricing trends observed in the primary dataset and to broaden the scope of the analysis across additional vehicle types.

Dataset Details: 1,000+ Rows & 10 Columns

Size: 160 kb

Access: Downloaded from Kaggle as a CSV file.

Source: Kaggle – <https://www.kaggle.com/datasets/ayeshasiddiqa123/cars-pre/data?utm>

Dataset 3: vehicle_price_prediction

An additional validation dataset focused on vehicle price prediction, providing a broader automotive pricing context to cross-reference findings from the primary sports car dataset.

Dataset Details: 1000000 Rows & 20 Columns (estimated)

Size: 28 MB

Access: Baselight/Kaggle

Source: https://baselight.app/u/kaggle/dataset/metawave_vehicle_price_prediction/table-explorer

2. Data Profiling

Dataset 1: sports_car_prices

Steps used to examine data quality:

- Apply Excel filter to the dataset to inspect each column individually.
- Check each column for null, blank, or missing values.
- Review unique values per categorical column (Car Make, Car Model) to detect inconsistencies in naming or capitalization.
- Inspect the Year column for out-of-range or unrealistic values.
- Inspect numerical columns (Engine Size, Horsepower, Torque, 0–60 MPH Time, Price) for outliers using min/max review and descriptive statistics.
- Identify any duplicate rows.

Notable Features:

- Multiple Car Makes represented (e.g., Porsche, Lamborghini, Ferrari, BMW, etc.)
- Year range spanning recent decades of sports car production.
- Performance metrics: Engine Size (L), Horsepower (HP), Torque (lb-ft), 0–60 MPH Time (sec).
- Price in USD: the dependent variable for regression analysis.

Recorded Inconsistencies:

- Some Car Model entries may have minor naming variations (e.g., spacing differences).
- Possible outliers in Price for ultra-premium/exotic models.
- 0–60 MPH Time may have missing values for certain models.
- LLM-generated data: may lack real-world variability and external pricing factors.

Datasets 2 & 3: car_price_analysis / vehicle_price_prediction

- Apply Excel filter and examine each attribute for inconsistency.
- Check for columns that overlap with the primary dataset (make, model, price, year) to enable cross-validation.
- Identify and document any missing values or formatting differences.
- Confirm column types match expected formats (numeric vs. string).

3. Data Wrangling

Dataset 1: sports_car_prices

- Remove duplicate rows, if any are identified during profiling.
- Standardize Car Make and Car Model column values. trim whitespace, fix capitalization inconsistencies (e.g., 'BMW' vs 'bmw').
- Handle missing values in the 0–60 MPH Time column: either impute using the median for each Car Make, or flag and remove records with missing performance data.
- Filter out records where Price is 0 or unrealistically low/high.
- Feature Engineer a 'Price Tier' column to bin Price (USD) into market segments.
- Feature Engineer a 'Performance Score' column by normalizing and combining Horsepower, Torque, and 0–60 MPH Time into a single composite metric for EDA.
- Feature Engineer a 'Decade' column extracted from the Year attribute to enable decade-level trend analysis.
- Rename all columns to use snake_case (e.g., car_make, engine_size_l, price_usd) for consistency.

Price Tiers:

- Under \$50,000
- \$50,000 – \$100,000
- \$100,000 – \$200,000
- \$200,000 – \$500,000
- \$500,000+

Datasets 2 & 3: car_price_analysis / vehicle_price_prediction

- Retain only columns relevant to cross-validation: make, model, year, price, and key performance specs.
- Standardize column naming to snake_case to align with the primary dataset schema.
- Remove records with missing price values.
- Filter to include only sports/performance vehicle categories where applicable.

4. Data Table Schema

Dataset 1: sports_car_prices

Field	Type	Description
car_make	STRING	The brand or manufacturer of the sports car (e.g., Porsche, Ferrari, Lamborghini).
car_model	STRING	The specific model or variant of

		the sports car (e.g., 911, Huracan, R8).
year	INTEGER	The production year of the vehicle. Format: YYYY (e.g., 2020).
engine_size_l	FLOAT	Engine displacement measured in liters (e.g., 3.8, 5.2). Indicates performance potential.
horsepower	INTEGER	Engine power output in horsepower (HP). Key driver of speed and performance.
torque_lb_ft	INTEGER	Rotational force produced by the engine in lb-ft. Affects acceleration and handling.
zero_to_sixty_sec	FLOAT	Time in seconds to accelerate from 0 to 60 MPH. Lower = faster (e.g., 3.2).
price_usd	FLOAT	Market price of the vehicle in US dollars. This is the dependent variable for analysis.
price_tier	STRING	Feature engineered. Price range bin: 'Under \$50K', '\$50K–\$100K', '\$100K–\$200K', '\$200K–\$500K', '\$500K+'.
performance_score	FLOAT	Feature engineered. Composite normalized score combining Horsepower, Torque, and 0–60 MPH Time.
decade	STRING	Feature engineered. Decade derived from the Year column (e.g., '2010s', '2020s').

Datasets 2 & 3: car_price_analysis / vehicle_price_prediction (Validation Datasets)

Field	Type	Description
car_make	STRING	Brand or manufacturer of the vehicle (standardized to match primary dataset).
car_model	STRING	Model name or variant of the vehicle.
year	INTEGER	Production year of the vehicle. Format: YYYY.
price_usd	FLOAT	Market price of the vehicle in US dollars. Used for cross-validation against the primary dataset.
engine_size_l	FLOAT	Engine displacement in liters, where available.
horsepower	INTEGER	Engine power output in horsepower, where available.